

Name: _____

MZUZU DIOCESE EXAMINATIONS

2017 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATION

AGRICULTURE

Monday, 27 March

Subject Number: M012/1

Time Allowed: 2 hours
(7.30-09.30am)

PAPER 1
(100 Marks)
(Theory)

Instructions:

1. This paper contains **12 pages**. Please check.
2. Answer all **ten** questions in **section A** and all **four** questions in **section B**.
3. The maximum number of marks for each answer is indicated against each question.
4. Answer all questions in the spaces provided.
5. Write your **name** and **class** on top of every page.
6. In the table provided on this page, tick against the question number you have answered.
7. Hand in your paper to the invigilator at the end.

Question Number	Tick if Answered	Do not write in these margins	
1			
2			
3			
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8			
9			
10			
11			
12			
13			
14			

Section A (60 marks)

Answer all the **ten** questions in this section

1. a. State any **two** ways of classifying weeds.

(2 marks)

- b. (i) Give any **one** reason why it is important to classify weeds accurately.

(1 mark)

- (ii) Describe any **one** cultural practice for controlling weeds.

(2 marks)

2. a. Mention any **two** beef breeds of cattle

(2 marks)

- b. State any **two** features that can be used to identify a beef breed of cattle

(2 marks)

Continued/...

c. State any **one** way in which each of the following are avoided in cross-pollination of crops.

- i. Accidental self-pollination

(1 mark)

- ii. Pollination by other foreign pollen.

(1 mark)

3. a. State any **one** importance of each of the following seed treatment procedures for pasture:

- i. Scarification

(1 mark)

- ii. Inoculation

(1 mark)

- iii. Pelleting

(1 mark)

b. Explain any **one** way in which each of the following affects seed rate of pasture during pasture establishment:

- i. Method of sowing

(2 marks)

Continued/...

ii. Seed size

(2 marks)

4. a. Explain any **one** importance of each of the following in agricultural experimentation

i. report writing

(2 marks)

ii. calculating averages

(2 marks)

b. State any **two** examples of environmental interference in agricultural experiments.

(2 marks)

c. Describe any **one** way of reducing environmental interference in agricultural experiments.

(2 marks)

Continued/...

5. a. State any **two** roles of estates in ensuring food security.

(2 marks)

- b. Explain any **two** ways in which farmers can achieve food self-sufficiency at household level.

(4 marks)

- c. State any **two** factors that help to expand international agricultural trade in Malawi.

(2 marks)

6. Form 4 students at Chule CDSS poured 260ml of soil in a measuring cylinder. The students also poured 140ml of water in the measuring cylinder. After 20 minutes they noted that the volume of soil and water was only 345ml.

- i. What was the aim of the experiment?

(1 mark)

Continued/...

- ii. Calculate the percentage of air contained in the soil sample above.

(2 marks)

7. **Table 1** shows farm gate and retail prices for some agricultural produce at Kalambwe Market. Use it to answer the questions that follow.

Produce	Farm gate price (MK/kg)	Retail Price (MK/kg)
Irish potatoes	350.00	700.00
Cabbages	80.50	180.75
Onions	600.00	750.00

- i. Calculate the marketing margins for each of the following crops:

(1) Irish potatoes

(2 marks)

(2) Onions.

(2 marks)

Continued/..

- ii. (1) Which produce in **(i)** has the smaller marketing margin?

(1 mark)

- (2) State any **one** possible cause of the difference in marketing margins in **(i)** above.

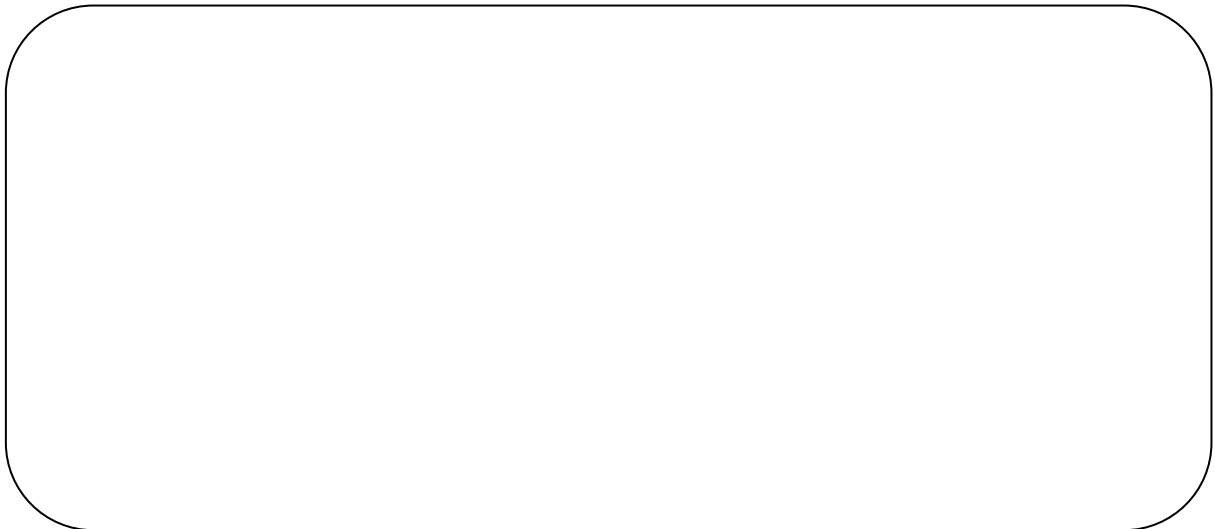
(1 mark)

8. A farmer bought a sprayer worthy K25,000.00 to be used for controlling pests in maize

- a. State **one** safety measure that should be observed when using a sprayer.

(1 mark)

- b. Assuming the sprayer has a life span of 10 years and a salvage value of K2, 000. 00, calculate the book value of the sprayer by the end of three years, using the straight line method.



(4 marks)

- c. State any **two** causes of depreciation of the sprayer calculated in **b.** above.

(2 marks)

Continued/...

9. A farmer noticed the following symptoms on his crops in a garden: scorched leaf margins, weak stalks resulting into lodging and small fruits, seeds and tubers.

- i. Identify the nutrient lacking in the crops.

(1 mark)

- ii. Name any **two** inorganic fertilizers that can be supplied by the farmer to correct the deficiency symptoms.

(2 marks)

- iii. State any **two** ways how the nutrient is depleted from the soil.

(2 marks)

10. a. State any **three** functions of roots.

(3 marks)

b. Explain any **one** reason why farmers are encouraged to plant deep rooted crops after shallow rooted crops when practising crop rotation?

(2 marks)

Continued/...

Section B (40 marks)

Answer all the **four** questions in an essay form except for question 12.

11. Explain any **five** roles of agro- based industries in supporting the growing population.

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(10 marks)

Continued/...

12. With the aid of a well-labeled diagram, describe the reproductive system of a cow.

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(10 marks)

Continued/...

13. Explain any **five** advantages of artificial insemination in livestock production.

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(10 marks)

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